

# DA5400W Tutorial 2

February 2, 2026

# Question 1

*Diagonalization & AM-GM Analysis*

DA5400W Tutorial 2

# Why Diagonalization?

- **Power Calculation:** Computes  $A^{100}$  instantly.
- **Data Science:** Fundamental to PCA (Principal Component Analysis).
- **Geometry:** Rotates coordinates to align with principal axes.

## The Roadmap

- 1 **Decompose**  $A$  into simpler parts ( $J + I$ ).
- 2 **Check AM=GM** to guarantee diagonalizability.
- 3 **Find Basis** vectors for the null space.
- 4 **Orthonormalize** using Gram-Schmidt.

# The Criterion: AM vs. GM

## Theorem

A matrix is diagonalizable  $\iff \mathbf{AM}(\lambda) = \mathbf{GM}(\lambda)$  for all  $\lambda$ .

### Algebraic Multiplicity (AM)

- How many times  $\lambda$  appears as a root of the characteristic polynomial.
- *"The Count"*

### Geometric Multiplicity (GM)

- The number of linearly independent eigenvectors for  $\lambda$ .
- Dimension of Null Space:  
 $\dim(\ker(A - \lambda I))$ .

# Generalization: The Decomposition Trick

**When can we use this?** For any  $n \times n$  matrix with diagonal  $a$  and off-diagonal  $b$ :

$$A = \begin{bmatrix} a & b & \cdots & b \\ b & a & \cdots & b \\ \vdots & \vdots & \ddots & \vdots \\ b & b & \cdots & a \end{bmatrix} = (a - b)I_n + bJ_n$$

The Eigenvalue Breakdown (Step-by-Step):

- 1 Start with  $J_n$ : Since it has Rank 1 and Trace  $n$ :

$$\lambda(J) = \{n, \underbrace{0, \dots, 0}_{n-1 \text{ times}}\}$$

- 2 Scale by  $b$ : Multiply eigenvalues by  $b$ :

$$\lambda(bJ) = \{bn, 0, \dots, 0\}$$

- 3 Shift by  $(a - b)I$ : Add  $(a - b)$  to every eigenvalue.

## Final Formula for Eigenvalues

- $\lambda_1 = bn + (a - b) = a + b(n - 1)$  (Appears once)
- $\lambda_{2 \dots n} = 0 + (a - b) = a - b$  (Appears  $n - 1$  times)

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# 1. Problem Setup

**Matrix  $A$  (Real & Symmetric):**

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

**Structural Insight (The Decomposition Trick):** Instead of calculating  $|A - \lambda I|$  directly, notice:

$$A = \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}}_{J_3 \text{ (Rank 1)}} + \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{I_3}$$

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## 2. Eigenvalues and AM

### Step A: Analyze $J_3$

- Rank = 1  $\implies$  Two eigenvalues are **0**.
- Trace = 3  $\implies$  Third eigenvalue is **3**.

Step B: Shift to  $A = J_3 + I$  Add 1 to the eigenvalues of  $J_3$ :

$$\lambda_A = \{0 + 1, 0 + 1, 3 + 1\} = \{1, 1, 4\}$$

### Algebraic Multiplicities (AM)

- $\lambda = 1$ : Appears twice  $\implies$  **AM = 2**
- $\lambda = 4$ : Appears once  $\implies$  **AM = 1**

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### 3. Geometric Multiplicity for $\lambda = 4$

For distinct eigenvalues, GM is always 1.

**Solve**  $(A - 4I)x = 0$ :

$$\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} x = 0 \implies x_1 = x_2 = x_3$$

**Eigenvector:**

$$v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \xrightarrow{\text{Normalize}} \boxed{u_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}}$$

**Check:**  $AM(4) = GM(4)$ . ✓

## 4. Geometric Multiplicity for $\lambda = 1$

**Solve**  $(A - 1I)x = 0 \implies J_3x = 0$ :

$$x_1 + x_2 + x_3 = 0$$

This is a plane through the origin. We need 2 basis vectors.

**Basis Selection:**

$$x_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad x_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

**GM Check:**

- We found 2 independent vectors.
- $\dim(\text{Null}) = 2$ .
- **GM = 2.**

**Verification Result**

Since  $\text{AM} = \text{GM} = 2$ , the matrix is fully diagonalizable.

## 4. Geometric Multiplicity for $\lambda = 1$

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- We found 2 independent vectors.
- $\dim(\text{Null}) = 2$ .
- **GM = 2.**

### Verification Result

Since  $\text{AM} = \text{GM} = 2$ , the matrix is fully diagonalizable.

## 5. The Orthogonality Problem

We have eigenvectors for  $\lambda = 1$ :

$$x_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad x_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

**Check Dot Product:**

$$x_1 \cdot x_2 = (-1)(-1) + (1)(0) + (0)(1) = \mathbf{1} \neq 0$$

### Goal

We need an **Orthonormal** set.

- 1 Normalize  $x_1$  to get  $v_1$ .
- 2 Use Gram-Schmidt to fix  $x_2$ .

## 6. Applying Gram-Schmidt

### Step 1: Normalize $x_1$

$$v_1 = \frac{x_1}{\|x_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

### Step 2: Orthogonalize $x_2$ Remove the component of $x_2$ parallel to $x_1$ :

$$w_2 = x_2 - \text{proj}_{x_1} x_2 = x_2 - \frac{\langle x_2, x_1 \rangle}{\langle x_1, x_1 \rangle} x_1$$

$$w_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.5 \\ -0.5 \\ 1 \end{bmatrix}$$

### Step 3: Normalize $w_2$

$$v_2 = \frac{w_2}{\|w_2\|} = \sqrt{\frac{2}{3}} \begin{bmatrix} -0.5 \\ -0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{-1}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}$$

## 6. Applying Gram-Schmidt

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$$v_2 = \frac{w_2}{\|w_2\|} = \sqrt{\frac{2}{3}} \begin{bmatrix} -0.5 \\ -0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{-1}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}$$

# Final Diagonalization

We have constructed the orthogonal matrix  $Q$ :

$$Q = [v_1 \ v_2 \ v_3] = \begin{bmatrix} \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

## The Spectral Theorem

Since  $A$  is symmetric:

- 1  $AM = GM$  (Diagonalizable)
- 2 Eigenvectors can be chosen Orthonormal ( $Q^T = Q^{-1}$ )
- 3  $A = Q\Lambda Q^T$

# Question 2

*Principal Component Analysis (PCA)*

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## Question 2: Problem Statement

Given the following 3D input data, identify the principal component.

|       | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ |
|-------|-------|-------|-------|-------|-------|
| Dim 1 | 1     | 2     | 3     | 4     | 5     |
| Dim 2 | 1     | 4     | 7     | 11    | 9     |
| Dim 3 | 9     | 6     | 4     | 4     | 2     |

Options:

(a)  $\begin{bmatrix} 0.9138 \\ -0.1035 \\ 0.3926 \end{bmatrix}$

(b)  $\begin{bmatrix} -0.2617 \\ 0.5891 \\ 0.7645 \end{bmatrix}$

(c)  $\begin{bmatrix} -0.4205 \\ -0.6223 \\ 0.2342 \end{bmatrix}$

(d)  $\begin{bmatrix} -0.3105 \\ -0.8014 \\ 0.5112 \end{bmatrix}$

# 1. Input Data (Question 2)

**Given 3D data points:**

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 9 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 3 \\ 7 \\ 4 \end{bmatrix}, \quad \mathbf{x}_4 = \begin{bmatrix} 4 \\ 11 \\ 4 \end{bmatrix}, \quad \mathbf{x}_5 = \begin{bmatrix} 5 \\ 9 \\ 2 \end{bmatrix}$$

**Goal:** Identify the principal component (the direction of maximum variance).

## 2. Step 1: Center the Data

Compute the mean ( $\mu$ ):

$$\mu_1 = \frac{15}{5} = 3, \quad \mu_2 = \frac{1 + 4 + 7 + 11 + 9}{5} = \frac{32}{5} = \mathbf{6.4}, \quad \mu_3 = \frac{25}{5} = 5$$

$$\mu = \begin{bmatrix} 3 \\ 6.4 \\ 5 \end{bmatrix}$$

Subtract mean from data ( $x_i - \mu$ ):

| Point | Dim 1 | Dim 2 | Dim 3 |
|-------|-------|-------|-------|
| $x_1$ | -2.0  | -5.4  | 4.0   |
| $x_2$ | -1.0  | -2.4  | 1.0   |
| $x_3$ | 0.0   | 0.6   | -1.0  |
| $x_4$ | 1.0   | 4.6   | -1.0  |
| $x_5$ | 2.0   | 2.6   | -3.0  |

### 3. Step 2: Compute Covariance Matrix

Formula:  $C = \frac{1}{n-1} \sum (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^T$  with  $n = 5$ .

$$C = \frac{1}{4} (X_{centered}^T X_{centered}) = \begin{bmatrix} 2.50 & 5.75 & -4.00 \\ 5.75 & 15.80 & -9.25 \\ -4.00 & -9.25 & 7.00 \end{bmatrix}$$

**Verification (Example Entry  $C_{12}$ ):**

$$\frac{(-2)(-5.4) + (-1)(-2.4) + (0)(0.6) + (1)(4.6) + (2)(2.6)}{4} = \frac{23}{4} = 5.75 \quad \checkmark$$

## 4. Step 3: Eigenvalues and Eigenvectors

Solving  $\det(C - \lambda I) = 0$  yields:

**Eigenvalues:**

**Eigenvectors (in columns):**

- $\lambda_1 = 0.1298$
- $\lambda_2 = 1.2415$
- $\lambda_3 = \mathbf{23.9287}$

$$V = \begin{bmatrix} 0.9138 & -0.1035 & \mathbf{-0.3105} \\ -0.2617 & 0.5891 & \mathbf{-0.8014} \\ -0.3105 & -0.8014 & \mathbf{0.5112} \end{bmatrix}$$

**Observation:** The third column corresponds to the largest eigenvalue ( $\lambda \approx 23.93$ ).

## 5. Final Solution

The Principal Component is the eigenvector associated with the largest eigenvalue ( $\lambda_{max} = 23.9287$ ).

$$\mathbf{v}_{PC} = \begin{bmatrix} -0.3105 \\ -0.8014 \\ 0.5112 \end{bmatrix}$$

### Answer Verification

Matches Option **(d)** from the tutorial sheet.

**Note:** In PCA, the sign of the vector does not affect the direction of the line (only the orientation), but here it matches the provided solution exactly.